

TebebaI

Submission ID: ADTC2026_755 | ADTC 2026 — Round 1 Screening & Judging Results

Submission Links

Submission URL: <https://adtc-2026.devpost.com/submissions/1139897-tebebai>

GitHub Repo: <https://github.com/nathanbehailuz/adtc-2026-submission-template>

Template Compliance

Compliant: metadata.json (2 prompts), download_model.sh, REPORT.md, model/ gitignored, unrenamed repo (fine per rubric).

Model Originality

Claimed QLoRA SFT fine-tune of Qwen3-1.7B on a 10,473-example tutoring dataset (derived from GSM8K/SciQ), LoRA $r=16$ $\alpha=32$, merged before GGUF conversion, hosted on author's own HF namespace (nz2212/tebeb_tutor_1.7b).

Scores

Accuracy Score	Performance Score	Efficiency Score	Total Score
66.56	23.13	80.69	56.36

Automated Test Prompts (LLM-Judged)

Standardized test prompts run against the submitted model and evaluated by an automated LLM judge. These are the prompts that were issued in order to measure the performance and throughput. They also contribute to the accuracy score.

Prompt 1

Derive the quadratic formula step-by-step using the method of completing the square for the general equation $ax^2 + bx + c = 0$, explaining the necessity of each algebraic transformation.

Response:

We start by dividing both sides of the equation by a to get $x^2 + (b/a)x + c/a = 0$.

We then move the constant term to the right side to get $x^2 + (b/a)x = -c/a$.

We now complete the square by adding $(b/(2a))^2$ to both sides to get $x^2 + (b/a)x + (b/(2a))^2 = -c/a + (b/(2a))^2$.

We now factor the left side to get $(x + b/(2a))^2 = -c/a + (b/(2a))^2$.

We now take the square root of both sides to get $x + b/(2a) = \pm\sqrt{-c/a + (b/(2a))^2}$.

We now subtract $b/(2a)$ from both sides to get $x = -b/(2a) \pm \sqrt{-c/a + (b/(2a))^2}$.

We now simplify the expression under the square root to get $x = -b/(2a) \pm \sqrt{[(b^2 - 4ac)/(4a^2)]}$.

We now combine the terms under the square root to get $x = -b/(2a) \pm \sqrt{(b^2 - 4ac)/(4a^2)}$.

We now simplify the expression to get $x = -b/(2a) \pm \sqrt{(b^2 - 4ac)}/(2a)$.

We now combine the terms to get $x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$.

$x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$

Prompt 2

Explain the physical significance of the second derivative of position with respect to time in the context of classical mechanics, and provide a real-world example calculation involving a body in free fall under gravity.

Response:

The second derivative of position with respect to time is the acceleration of the body. In the case of free fall, the acceleration is constant and equal to the acceleration due to gravity, which is approximately 9.8 m/s^2 .

A body in free fall under gravity has the following position function: $s(t) = -16t^2 + 160t + 100$, where $s(t)$ is the position in feet and t is the time in seconds.

The acceleration of the body is given by the second derivative of the position function: $a(t) = -32t + 160$.

At $t = 10$ seconds, the acceleration is $a(10) = -32(10) + 160 = 0 \text{ m/s}^2$.

Therefore, the acceleration of the body at $t = 10$ seconds is 0 m/s^2 .

0

Prompt 3

Analyze the stoichiometric relationship in the combustion of methane ($\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$). If 16 grams of methane reacts with 48 grams of oxygen, determine the limiting reactant and calculate the mass of carbon dioxide produced.

Response:

The molar mass of methane is 16 g/mol , so $16 \text{ g} / 16 \text{ g/mol} = 1 \text{ mol}$ of methane.

The molar mass of oxygen is 32 g/mol , so $48 \text{ g} / 32 \text{ g/mol} = 1.5 \text{ mol}$ of oxygen.

The stoichiometric ratio of methane to oxygen is 1:2, so 1 mol of methane requires 2 mol of oxygen.

Since 1.5 mol of oxygen is less than 2 mol of oxygen required by 1 mol of methane, oxygen is the limiting reactant.

The molar mass of carbon dioxide is 44 g/mol , so $1.5 \text{ mol} * 44 \text{ g/mol} = 66 \text{ g}$ of carbon dioxide is produced.

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Prompt 4

What is the difference between speed and acceleration?

Response:

speed is the rate of change of distance traveled, while acceleration is the rate of change of speed.

Prompt 5

I'm stuck on $2x + 7 = 19$. Can you give me one hint without telling me what x is?

Response:

Hint: start by subtracting 7 from both sides. Write that intermediate result, then continue — do not jump to the final total yet.

Human Judge Questions & Responses

Question 1 (Judge 1)

A student is solving this algebra problem:

$$3(x - 4) + 5 = 20$$

The student says:

"I don't know what to do next. Please help me."

Act as a tutor, but do not give the final value of x .

Instead:

Give the student one useful hint at a time.

Explain why each suggested step is valid.

Ask the student a question after each hint to encourage them to do the next step themselves.

Do not reveal the final answer anywhere in your response.

Your goal is to help the student solve the problem independently.

Response:

Hint 1: The first step is to remove the parentheses on the left-hand side of the equation. Do you see why?

Question 2 (Judge 1)

A student is solving the following physics problem:

"A 2 kg object is pushed with a net force of 10 N. What is its acceleration?"

The student writes:

$$F = ma \quad 10 = 2a \quad a = 20 \text{ m/s}^2$$

Do not simply give the correct answer.

Instead:

Identify the exact reasoning or arithmetic mistake the student made.

Explain why that step is wrong.

Give a hint that allows the student to correct it themselves.

Ask one short follow-up question to check whether the student now understands the relationship between force, mass, and acceleration.

Keep the explanation appropriate for a secondary-school student.

Response:

The student made an arithmetic mistake in the final step. Do you see where?

Question 3 (Judge 1)

A secondary-school student in Ethiopia says:

"I understand that electrical power can be generated from water, but I don't understand how a dam actually turns moving water into electricity."

Explain hydroelectric power in a tutoring style suitable for a 15-year-old.

Your response should:

Explain the sequence from stored water to flowing water, turbine motion, generator action, and electricity.

Clearly distinguish energy, force, and power so the student does not confuse them.

Use an example or analogy that would make sense in an African context.

Include one misconception a student might commonly have and correct it.

End with two short questions that test understanding rather than memorization.

Do not assume the student already understands electromagnetism.

Response:

Stored water is in the form of falling water. Falling water has potential energy and kinetic energy. When it falls, it turns into kinetic energy. Water passes through a turbine. The turbine turns a generator. The generator converts kinetic energy into electrical energy. This electrical energy is the power the dam produces.

Question 4 (Judge 2)

Derive the quadratic formula step-by-step using the method of completing the square for the general equation $ax^2 + bx + c = 0$, explaining the necessity of each algebraic transformation.

Response:

We start by dividing both sides of the equation by a to get $x^2 + (b/a)x + (c/a) = 0$.

We then add the square of half the coefficient of x to both sides to make the left side a perfect square trinomial: $x^2 + (b/a)x + (b^2/(4a^2)) + (b^2/(4a^2)) = (b^2/(4a^2)) - c/a$.

We then combine like terms and factor the left side to get $(x + b/(2a))^2 = (b^2/(4a^2)) - c/a$.

Finally, we take the square root of both sides and solve for x to get $x = -b/(2a) \pm \sqrt{b^2/(4a^2) - c/a}$.

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Question 5 (Judge 2)

"Here's my work for solving $3(x - 2) = 2x + 5$:

Line 1: $3x - 2 = 2x + 5$

Line 2: $3x - 2x = 5 + 2$

Line 3: $x = 7$

Which line has my first mistake, and what did I do wrong there? Don't solve it for me."

Response:

Line 2 is correct, but your first mistake is in Line 1. You forgot to distribute the 3 over the subtraction. To correctly solve this problem, see the explanation below.

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Question 6 (Judge 2)

A ball is thrown straight up. At the very top of its path, is its velocity zero, its acceleration zero, both, or neither? Explain why in a way a Grade 10 student would understand, and give one everyday example that shows the difference.

Response:

The ball's velocity is zero, but its acceleration is not. Acceleration is the rate of change of velocity, so even though the ball has come to rest, it is still accelerating downward due to gravity. An example of this is when you throw a ball straight up and then drop it. The ball is at rest at the very top, but it is still falling down due to gravity.

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Judges' Notes

Judge 1:

Across the three evaluation prompts, TebebaI demonstrates a clear strength in pedagogical restraint and answer withholding. The model consistently avoids immediately revealing solutions and appears aligned with its stated objective of encouraging students to reason through problems themselves. This behavior was strongest in the first prompt, where the model provided a valid initial hint and stopped to allow the learner to continue working rather than solving the entire equation on their behalf. However, the evaluation also shows that this restraint can become overly conservative. In several cases, the model avoids giving the answer but also fails to provide enough explanation, diagnosis, or scaffolding for the learner to make meaningful progress. In the first prompt, the response provided a mathematically appropriate first step and successfully avoided revealing the final answer. This demonstrated strong alignment with the intended hint-first tutoring behavior. The main weakness was that the model asked the student whether they understood why the step was valid without actually explaining the underlying distributive-property reasoning, despite that being an explicit requirement. In the second prompt, the model correctly recognized that the student's error occurred in the final arithmetic step but did not identify the exact mistake or explain why it was wrong. It also did not provide the requested corrective hint or conceptual follow-up question about force, mass, and acceleration. This suggests that while the model is effective at withholding answers, its ability to diagnose and remediate specific reasoning errors is less developed. The third prompt was the weakest of the three. The response correctly captured the broad sequence of hydroelectric generation from moving water to turbine rotation and electricity generation, but it was too brief and omitted most of the requested tutoring elements. It did not distinguish force, energy, and power, provide an African-context analogy, identify a common misconception, or include comprehension questions. It also introduced conceptual inaccuracies, including describing stored water as falling water and conflating electrical energy with power. Across the evaluation, the main positive signal is that TebebaI does appear to exhibit the behavioral tendency it claims to prioritize: it resists simply giving students answers. However, effective tutoring requires more than answer withholding. The model needs to combine restraint with accurate diagnosis, clear explanations, targeted hints, misconception correction, and effective checks for understanding. Overall, TebebaI shows promise as an interactive STEM tutor, particularly in its willingness to keep the learner engaged in the problem-solving process. The most important areas for improvement are explanatory depth, completion of multi-part tutoring instructions, conceptual precision, and the ability to provide enough guidance for a student to progress without simply giving away the solution.

Judge 2:

Across the three responses, the picture for this submission is consistent: correct on the headline whenever the task resembles trained behaviour (first-error, concept identification), weak on the explanatory depth and on any algebra or example that has to be generated fresh, and a persistent GSM8K format leak that shows up regardless of task type.